

Estimating the dynamics of greenhouse gas emission during black soldier fly larvae growth under controlled environmental conditions

Giacomo Rossi^a, Shikha Ojha^{a,b}, Werner Berg^a, Werner B. Herppich^a, Oliver K. Schlüter^{a,c,*}

^a Department of Systems Process Engineering, Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Max-Eyth-Allee 100, 14469, Potsdam, Germany

^b Department of Land Sciences, School of Science and Computing, South East Technological University, Cork Road, X91 KOEK, Waterford, Ireland

^c Department of Agricultural and Food Sciences, University of Bologna, Piazza Goidanich 60, 47521, Cesena, Italy

ARTICLE INFO

Handling Editor: Jian Zuo

Keywords:

Edible insects
Insect rearing
GHG
Carbon dioxide
Gas exchange
Novel food

ABSTRACT

Edible insects are considered a sustainable alternative protein source. Several studies have indicated that greenhouse gas (GHG) emissions from insects are generally low. These data were usually recorded in closed static chambers, which did not reflect the actual rearing conditions. On the other hand, open respiration chambers need high amounts of substrate and insects, making the analyses time consuming and expensive. In the present study, a novel small-scale open gas exchange recording system for evaluation of GHG emissions from insects is presented. The device consists of four independent respiration chambers, each one equipped with a pump for supplying air, a flow meter and a septum for sampling the flowing air. A validation experiment was carried out using black soldier fly larvae as model organisms and moistened chicken feed (dry matter 40%) as rearing substrate. Three independent runs were conducted, each with two replicates, using 200 neonates larvae and 300 g of substrate. Moistened chicken feed (300 g) without larvae was used as control. Throughout the 12 days rearing period, gas sampling occurred every 12 h. Data interpolation considering a linear tendency between two successive sampling points was applied and overall amounts of gases were quantified as area under the curve. Irrespective of the run, CO₂ emissions in the insect-substrate systems increased during the initial 7 d of experiment, consistently followed by a rapid decrease. Multiple cross correlation analysis showed high reproducibility ($r \geq 0.9$), while significant differences in the CO₂ emissions between insects-substrate system (range 7.76–11.88 g g⁻¹ dry insects) and sole substrate (range 0.20–0.33 g g⁻¹ initial dry substrate) were observed. Very limited N₂O (≤ 1.13 mg g⁻¹ dry insects) and CH₄ (≤ 1.33 mg g⁻¹ dry insects) emissions were detected in this experiment, indicating that the performed black soldier fly larvae rearing did not result in significant production of these GHGs. Overall, the experiment showed that the novel device may be particularly suited for quick and reliable estimation of GHG emissions from edible insects.

1. Introduction

Approximately 30% (15.8 Gt CO₂-eq) of the overall (52.5 Gt CO₂-eq) global greenhouse gas (GHG) emissions are generated by the agro-food system (Li et al., 2022). Excluding the non-food agriculture, the food supply chain alone is responsible for 13.7 Gt emissions (Poore and Nemecek, 2018), with 7.4 Gt of CO₂-eq originating within the farm gate (crops and livestock production) (FAO, 2022). Emissions from meat production are estimated to be approx. 2.8 Gt of CO₂-eq per year, accounting for 39% of the overall agriculture GHG emissions (excluding transport and land use) (Li et al., 2022). A large part of such emissions is shared among ruminant animals with cattle and sheep responsible for 30

and 24 kg of CO₂-eq per kg of meat, respectively (FAO, 2022). Hence, it is evident that the modern food system is unsustainable and requires transformation.

Novel and future food sources have been described as perfect means for ensuring the ecological transition of the food system while safeguarding the nutritional quality of the diet (Parodi et al., 2018). Whether integrated with other novel food sources, inclusion of edible insects in the daily diet may represent the turning point toward a sustainable food system (Mazac et al., 2022).

Research conducted to date has demonstrated that insects can efficiently grow on waste, by-products and decaying materials while emitting very low GHG (Moruzzo et al., 2021). Despite insect rearing is

* Corresponding author. Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Max-Eyth-Allee 100, 14469, Potsdam, Germany.

E-mail address: oschluter@atb-potsdam.de (O.K. Schlüter).

<https://doi.org/10.1016/j.jclepro.2024.143226>

Received 20 November 2023; Received in revised form 12 July 2024; Accepted 21 July 2024

Available online 23 July 2024

0959-6526/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

typically associated with low gas emissions (Boakye-Yiadom et al., 2022), only a few studies investigating the GHG emissions of edible insects have been published so far. Moreover, with only few exceptions (Parodi et al., 2020), most of these studies investigated the gas emissions in closed static chambers (Ermolaev et al., 2019; Mertenat et al., 2019). Although such systems can be useful for shaping a first idea on the insects' gas emissions, they cannot be considered entirely accurate since they do not simulate the actual insect growing conditions. Accumulation of CO₂ and moisture, and depletion of O₂ within such chambers may indeed lead to metabolic modifications, which, concomitantly, affect gas emissions (Akbar et al., 2016; Boakye-Yiadom et al., 2022; Mertenat et al., 2019). On the other hand, open systems (Matussek and Herppich, 2020) may be difficult and costly to operate, needing huge amounts of substrate and insects to detect very low concentrations of relevant gases (Parodi et al., 2020). Therefore, in order to measure insects' gas emissions efficiently and cost-effectively, to compare various substrates and to offer important information to entrepreneurs and policy makers, small scale and efficient gas emissions monitoring systems need to be developed. The aim of this study was to design and test a new emission chamber for efficiently recording the GHG emissions of insects, while simulating real rearing conditions. This paper presents and describes the new device in the first part. In the second section, results of a validation experiment are discussed. Black soldier fly larvae (BSFL) and moistened chicken feed were used as model organisms.

2. Description of the gas emission chamber

The innovative climate-controlled gas exchange system (Fig. 1a) for continuous recording of insects' gas emissions was based on a small-scale respiration chamber for measuring the metabolic activity of fresh vegetables (Linke et al., 2010; Rux et al., 2017). The new system consisted of four acrylic glass cylinders (gas exchange chamber, V = 9.42 L, H = 30 cm, Ø = 20 cm, Fig. 1b) in which boxes with substrate and insects could be housed. Each cylinder could be hermetically sealed by a removable front acrylic glass push-fit cap (Ø = 22 cm, W = 1 cm) fitted with polyurethane frame closure connection (2.5 cm height). The closing system consisted of a rectangular aluminium rod (H = 3 cm, L = 24.5 cm, W = 1 cm) and a polyurethane-screened aluminium cylinder (Ø = 5.5 cm, W = 1 cm). In each chamber, three sensors recorded temperature, relative humidity and air pressure ((d), ALMEMO FHAD 46-C2 multi-sensor module, Ahlborn Mess-und Regelungstechnik GmbH, Holzkirchen, Germany). Sensors, placed along the vertical central line of each unit and fixed through screw IP68 polyamide Cable Glands connectors (Conrad Electronic SE, Hirschau, Germany), were connected to a PC. Data were recorded at 1 min interval and analysed

with the help of an Almemo-COM-based program (written by C. Regen, ATB, Potsdam, Germany).

Continuous airflow through each chamber was obtained by an inlet-outlet system. The inlet system (Fig. 1, green pipes (a)), consisting of PVC tubes (inside diameter (iØ) = 1.5 cm), led temperature and humidity conditioned air from a climate chamber into a PVC sampling box (H = 19 cm, L = 10.5 cm, W = 12 cm) equipped with a polyisoprene 11 mm septum (Sigma-Aldrich, St. Louis, Missouri, USA) for gas sampling and further to a second central input PVC box (H = 19 cm, L = 10.5 cm, W = 12 cm). Through smaller silicon tubes (iØ = 1.0 cm), four pumps ((b), Daypower Luftpumpe LP36-12, Pollin Electronic GmbH, Pförring, Germany) pushed the inlet air into each gas exchange chamber (c). Flow rate of each inlet line was independently controlled and regulated by adjusting the voltage of the pump.

For the outlet system (Fig. 1a, red pipes (e)) 1.0 cm-silicon tubes (overall length 210 cm) were placed on the front cap of each chamber and fixed through a 4 cm long polyamide semi-flexible pipe (Landefeld Druckluft und Hydraulik GmbH, Kassel, Germany). At 200 cm, a Rotilabo T connector (Carl Roth GmbH & Co. KG, Karlsruhe, Germany), provided with a natural polyisoprene 11 mm septum ((g), Sigma-Aldrich, St. Louis, USA) was installed for gas sampling. An analogue flowmeter ((f), placed on the final section of the outlet tube was finally responsible for recording airflow rates.

In order to ensure exact temperature control, the entire system was placed inside a climate chamber (Bio Line VB1014, Vötsch Industrietechnik GmbH, Balingen, Germany) from which the inlet air was conditioned. Besides PC and power supply, flowmeters, inlet sampling box and outlet tubes including the outlet sampling septa were placed outside the climate chamber.

3. Materials and methods

3.1. Substrate and insects

Commercial chicken feed (all-mash A Mehl M.Cocc., Deuka companion GmbH, Düsseldorf, Germany) with an initial dry matter content of $91.62 \pm 0.07\%$ was ground (10000 rpm) into a fine powder with a knife mill (Retsch GmbH, Hann, Germany), applying three cycles of 10 s each, separated by breaks of 30 s to homogenise the bulk. Aliquots (133.5 g) of this powder were transferred into a transparent 2 L polypropylene box (Ikea Deutschland GmbH & Co. KG, Munich, Germany) and moistened at approx. 60% of humidity by adding 166.5 g of distilled water (final substrate mass 300 g). To avoid excess of free water on the surface, prepared substrates were kept two days (48 ± 2 h) at 10 °C before adding larvae. For each run, four boxes with substrate were

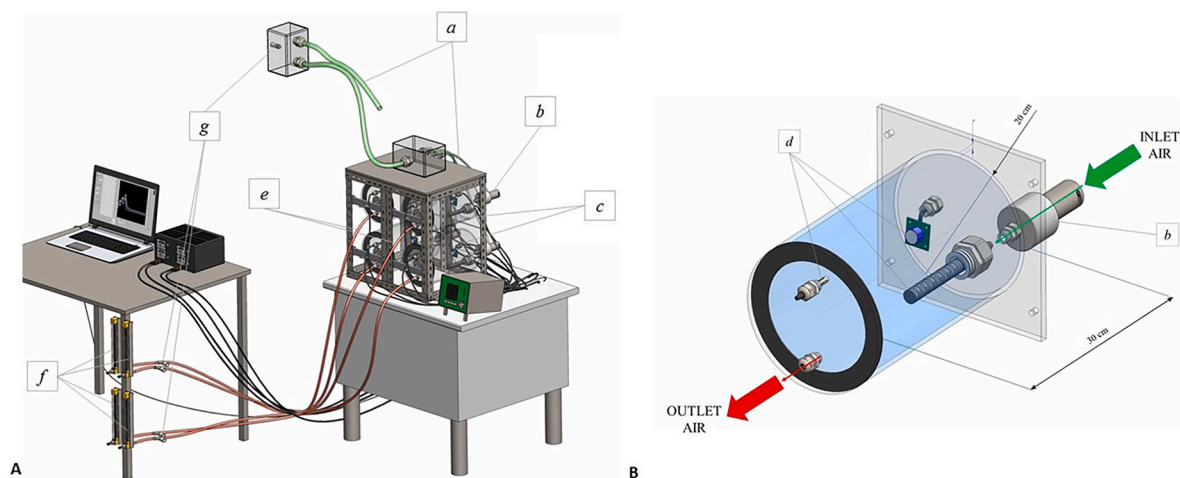


Fig. 1. Overview of the gas exchange system for online recording of insects' GHG emissions (A) and overview of the specification of the gas exchange chamber (B). SolidWorks 3D CAD 2021 professional design software was employed for drawing the device.

prepared but insects were added only in two boxes (treatments: T1, T2), while the other two boxes were used as control (C1, C2).

Black soldier fly larvae (BSFL, *Hermetia illucens*) were chosen as model insects. Neonates BSFL (less than 12 h old), generated from eggs (Hermetia Baruth GmbH, Baruth/Mark, Germany) laid on corrugated cardboard were used. Eggs were left to hatch without substrates in a polypropylene transparent box placed in a climate chamber ($30 \pm 1^\circ\text{C}$, $70 \pm 5\%$ RH, 0:24 L:D). Eggs eclosion started approximately 36 h after receiving the eggs and only the larvae originated in the following 12 h (36–48 h from the purchase) were used in the experiment. Exactly 200 neonate larvae were carefully transferred in two of the four boxes with substrate through a camel brush. All the boxes were closed with a polypropylene transparent lid (Ikea Deutschland GmbH & Co. KG, Munich, Germany) equipped with a rectangular hole ($15\text{ cm} \times 10.5\text{ cm}$) screened with polyester black mosquito net ($1\text{ mm} \times 1\text{ mm}$ mesh, HaGa-Welt GmbH & Co. KG, Nordstemmen, Germany). Number of BSFL and amount of feed were carefully defined with the aim of ensuring approximately 125 mg of fresh feed/larvae/day or 50 mg of dry feed/larvae/day as suggested by [Diener et al. \(2009\)](#) and [Dzepe et al. \(2021\)](#). After adding larvae, all boxes were immediately transferred into individual gas exchange chambers, placed inside a Bio Line climate chamber ($27 \pm 1^\circ\text{C}$ and $70 \pm 5\%$ RH). Each gas exchange chamber was hermetically closed, drain pumps switched on and data recording initialized. No further feed was provided during the experiment.

3.2. Experimental set-up and gas emission measurements

Experiments were repeated three times (three runs). Two independent replicates were performed for each run (six replicates in total).

Actual concentration of N_2O , CH_4 and CO_2 emitted by insects and substrates were discontinuously measured through GC analysis of the air sampled from the inlet and outlet systems. Gas sampling was carried out every 12 h (every day at 8:00 a.m. and 8:00 p.m. ($\pm 30\text{ min}$)), along the entire insect rearing period of 12 d. At each sampling point, three gas samples were sampled with a 25 mL gastight glass syringe (1025 TLL, PTFE Luer Lock, Hamilton Company, Nevada, USA) equipped with a Sterican Gr2-G21 $0.8\text{ mm} \times 40\text{ mm}$ needle and immediately conveyed in a pre-evacuate (2.7 mbar of negative pressure) 25 mL glass vial equipped with a PTFE/butyl rubber septum stopper.

Sampling started approximately 10 h after setting up the box inside each gas exchange unit (stabilization period = $10 \pm 1\text{ h}$). CO_2 and O_2 concentrations, relative humidity and pressure inside each unit were monitored before and after each sampling and the airflow rates were adjusted to keep the relative humidity inside the unit between 65 and 75% RH.

To minimise the interaction with external environment, gas exchange units and climate chamber were not opened during the experiments. After 12 d, all gas exchange chambers were opened and the rearing boxes were collected. BSFL were harvested from T1 and T2 treatments by handpicking them with forceps, carefully cleaned with towel paper, counted and weighted on an analytical balance (CPA224S-OCE, Sartorius AG, Göttingen, Germany). Mass of residuals (frass from T1 and T2, substrate residuals from C1 and C2) was also recorded.

3.3. Gas quantification

Air samples were analysed every Monday, Wednesday and Friday morning. When needed, air samples were dark-stored in a BD-S 115 safety cabinet (Binder GmbH, Tuttlingen, Germany) at 20°C to minimise any possible external impact. Air samples were analysed with a GC-2014 gas chromatograph (Shimadzu Deutschland GmbH, Duisburg, Germany) equipped with FID and ECD detectors as well as a HS-20 auto sampler. Four different Agilent stainless steel packed columns (Haysep N, 1.0 m; Haysep D, 4.0 m; Mole Sieve 5A, 2.5 m; Haysep N, 1.5 m) were used (overall length 9 m, inner diameters 2 mm and mesh size 80/100). Nitrogen was used as carrier gas, while hydrogen and synthetic gas

Alphagaz 1 (Air Liquide Deutschland GmbH, Berlin, Germany) were the fuel gasses for FID.

Air samples (2 mL) from the vials were directly injected into the GC. Gas flow was set at 25 mL min^{-1} , oven temperature at 80°C , and FID and ECD temperatures at 250°C and 325°C , respectively. CH_4 concentration (detection range: 1–100 ppm) was measured directly from the sample, while CO_2 was converted to CH_4 within a methaniser operating at 380°C , prior detection by FID (detection range: 1–100000 ppm). N_2O concentration was finally measured by ECD (detection range: 0.05–100 ppm). Effective gas concentration (ppm) was calculated by the LabSolutions LCGC software (Shimadzu Deutschland GmbH, Duisburg, Germany) using four standard calibration gas mixtures (Air Liquide Deutschland GmbH, Düsseldorf, Germany). For each studied gas, intra-run accuracy of the GC was computed as average of the absolute difference between the recorded value and the average of three independent determinations performed on the same standard mixture.

3.4. Chemical analyses of substrates, larvae and residuals

Dry matter (DM), volatile solids (VS), ash, total nitrogen (N), total carbon (C), total organic carbon (TOC), total sulphur (S) contents, and gross energy were analysed on substrates, 12 d old-insects and residuals. Additionally, ammonium ($\text{NH}_4\text{-N}$), nitrate ($\text{NO}_3\text{-N}$) and nitrite ($\text{NO}_2\text{-N}$), neutral detergent fibre (NDF), acid detergent fibre (ADF) and acid detergent lignin (ADL) contents were determined in substrates and residuals. Insects were analysed after lyophilisation, while substrates and residues after oven-drying at 60°C for 48 h. DM and ash contents were determined gravimetrically after oven-drying at 105°C overnight (NFTA 2.1.4 official moisture method) and incineration at 550°C for 4 h (DIN 38414-EN 12879), respectively. VS content was calculated by difference between DM and ash. Total N content was determined with the Kjeldahl method (AOAC, 2000-method 992.15), while C and S were analysed with the element analyser VARIO EL (Elementar Analysensysteme GmbH, Langenselbold, Germany), following the producer's instructions. TOC was gravimetrically determined by quantifying the amount of carbon released after sample combustion in acidic environment (DIN-EN 13137:2001–12), while total protein content was computed by multiplying the N content with 4.76 for insects ([Janssen et al., 2017](#)) and 5.70 for substrates ([Sriperum et al., 2011](#)). Gross energy was determined on the samples dried at 105°C for 24 h, following the oxygen bomb method (ISO 9831:1998), while NDF, ADF, ADL contents were analysed according to the ISO 13906:2008 and AOAC 973.18 methods, using the Ankom 2000 Fiber Analyser (Ankom Technology, Macedon, USA). Finally, $\text{NH}_4\text{-N}$, $\text{NO}_3\text{-N}$ and $\text{NO}_2\text{-N}$ contents were quantified with the ion chromatography system Dionex ICS-1000 (Thermo Fisher Scientific, Waltham, USA) after samples extraction with distilled water ($\text{NO}_2\text{-N}$) or 0.0125 M CaCl_2 solution ($\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$). Chemical determinations of substrates were performed in triplicate, while insects and residues were analysed in duplicate. In both cases, results from the same sample were averaged.

3.5. Calculations, data modelling and statistical analysis

Larvae growth and substrate reduction performances of insects were computed through the following indices:

- (1) Bioconversion rate (BCR; %) = (total final larvae mass (g)/total substrate mass (g)) * 100
- (2) Assimilated feed (g DM) = total substrate dry mass (g) – total frass dry mass (g)
- (3) Feed conversion rate (FCR) = (assimilated feed (g))/(final larvae dry mass (g))
- (4) Substrate reduction (%) = (assimilated feed (g)/substrate dry mass(g)) * 100
- (5) Larval survival (%) = (final larval number/initial larval number) * 100

As the experiments started with 12 h old-larvae, initial larval mass was assumed to be zero.

Differences in gas concentrations (outlet – inlet) were multiplied by flow rates (L min^{-1}) and modelled through interpolation between subsequent data points. The overall volume (mL) of each gas emitted from each single chamber was estimated by calculating the area under the curve. Computed volume of each gas was then converted into mass (g) by multiplying it with the ratio of molar mass of the respective gas (CO_2 : $44.009 \text{ g mol}^{-1}$; N_2O : $44.013 \text{ g mol}^{-1}$; CH_4 : $16.043 \text{ g mol}^{-1}$) and the molar volume of an ideal gas (22.41 L mol^{-1}). Intra run and inter run reproducibility were therefore computed through multiple cross correlation analysis.

Two-samples *t*-tests ($P < 0.05$) with observations nested within the same run was applied for comparing the overall amount of each gas emitted by the system substrate + insects with the amount of gas emitted by the sole substrate. Additionally, in order to evaluate if gas concentration in outlet and inlet air were effectively different, differences in gas concentration were compared with the accuracy of the GC. Differences were therefore considered reliable only if they exceeded the accuracy of the measuring device.

One-sample *t*-test was finally used to evaluate whether emissions of N_2O and CH_4 were significantly different from zero.

Data modelling, interpolation and area under the curve procedure were performed in MATLAB R2021b (The MathWorks Inc., Natick, MA, USA), while statistical analyses were conducted on R software version 4.2.3 (R-CoreTeam, 2023; Vienna, Austria).

Chemical data of substrates, insects, frass (consisting in residual substrate, larvae excreta and larvae exuviae from treatments T) and residues (substrate residual from treatments C) were used for computing the absolute mass (g) of each component, which were applied for

calculate the overall mass balances as described by Parodi et al. (2020). To this end, the mass of each element emitted as gas (g) was calculated by multiplying the overall mass of the gas (g) by the ratio of molar mass of the considered element ($12.011 \text{ g mol}^{-1}$ for C and $14.007 \text{ g mol}^{-1}$ for N) and the molar mass of the respective gas. Element distribution between larvae, residual and air as well as percentage of not recovered element (element lost) were computed through the following equations (Parodi et al., 2021):

- (6) Element in larvae (%) = (element in larvae (g)/element in substrate (g)) * 100
- (7) Element in residual (%) = (element in residual (g)/element in substrate (g)) * 100
- (8) Element in gas (%) = (element in emitted gas (g)/element in substrate (g)) * 100
- (9) Element lost (%) = $100 - (\text{element in larvae (\%)} + \text{element in residual (\%)} + \text{element in gas (\%)})$

4. Results and discussion

The aim of this investigation was to test and validate the new gas exchange chamber. Three independent runs and two replicates per run evaluating the GHGs emitted by BSFL reared on moistened chicken feed were carried out. Therefore, emissions of the system substrate + insects (treatment T) were compared with the emission of the sole substrate (treatment C), and experimental reproducibility was evaluated through a multiple cross correlation analysis.

Although GHG emissions from insects are known to be highly variable (Parodi et al., 2020), the recovered data showed high inter-run reproducibility, suggesting suitability of the novel device for rapid and

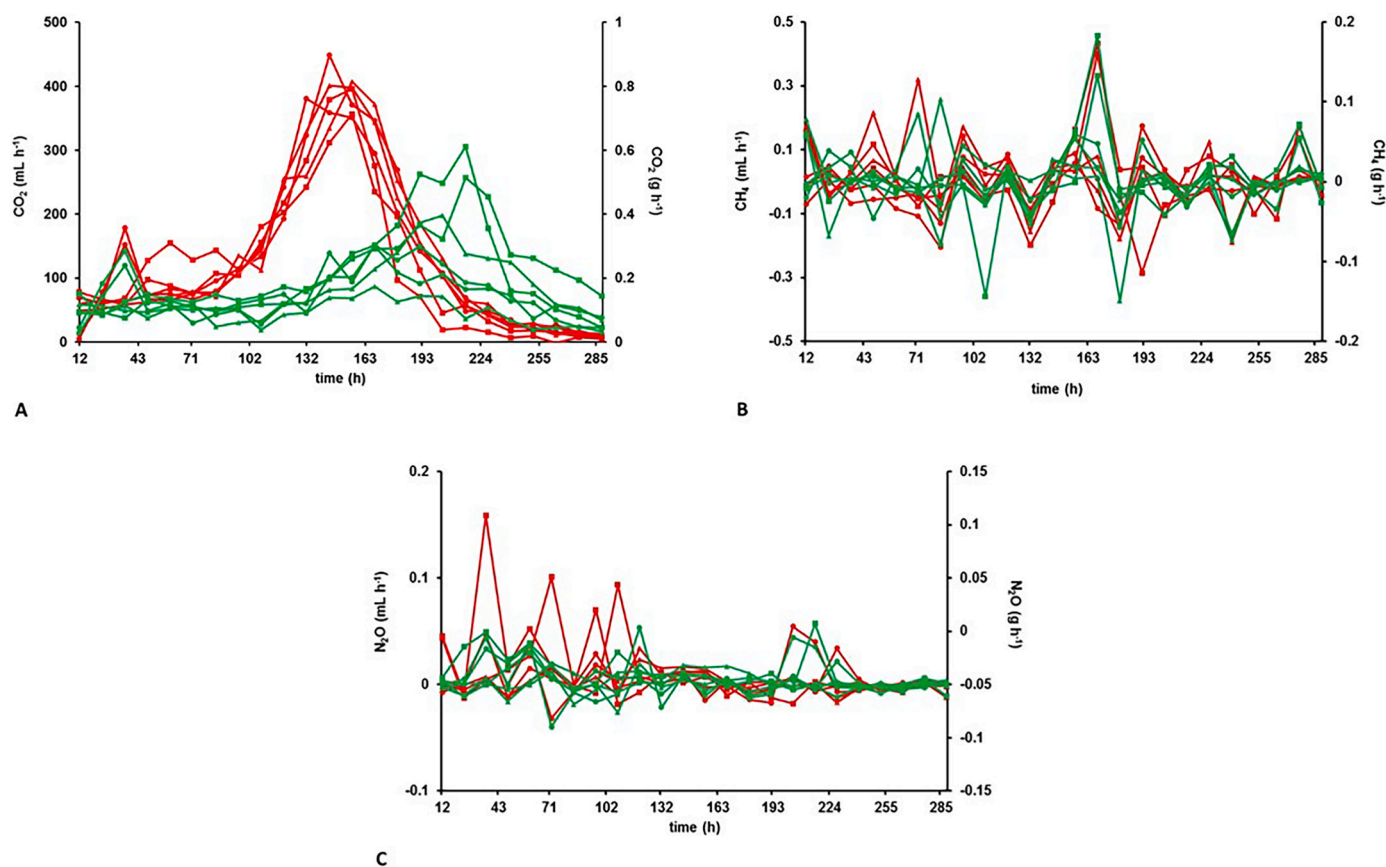


Fig. 2. Gas emission curves for CO_2 (A), CH_4 (B), N_2O (C) recorded during the experimental runs. Red curves represent the gas emitted by the substrate + insects systems (treatments T) while green curves indicated the gas emitted by the sole substrate (treatments C). Different markers are used for different run: RUN-1 = square, RUN-2 = round, RUN-3 = triangle. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

reliable estimation of insect gas emissions.

4.1. GHG emissions trends and experimental reproducibility

CO₂ emissions from the treatments T (Fig. 2A, red curves) were low at the beginning, increased after approximately 3.5 d, peaked between day 5 and day 7 and decreased again up to days 8–9. Such bell-shaped trend was observed in all the runs, with very good overlapping of curves. Cross correlation analysis (supplementary material S1) displayed a strong significant positive intra and inter run correlation ($r \geq 0.9$) between treatments T, which suggested high reproducibility of the process. The aforementioned dynamics have already been noticed by other authors (Bekker et al., 2021; Ermolaev et al., 2019; Pang et al., 2020a; Perednia et al., 2017), further supporting the validity of the measuring procedure. The main explanation for this particular dynamic might be found in the activity of BSFL and in the nature of the rearing substrate. CO₂ is mainly originated from microbial and larval metabolism (Boakye-Yiadom et al., 2022) whereby it is plausible to expect low emissions at the beginning, followed by a fast increase and a final reduction and stabilization. Bekker et al. (2021) also observed that with a substrate moisture of 65–75%, larvae started producing significant CO₂ only after reaching 5–6 mg of mass. After this point, linear correlation between CO₂ production and larval mass was usually observed (Laganaro et al., 2021). In the present experiment, larvae less than 12 h old were used. Consequently, they could have had a very low activity during the first hours of experiment, explaining the low CO₂ emission observed at the beginning. As the larvae grew, their metabolic activity (i. e., respirational activity) increased (Gligorescu et al., 2019), resulting in higher CO₂ release. Similarly, the reduction of CO₂ release observed at the end of each run could suggest a decline in larvae metabolic activity, mainly due to the achievement of the critical mass (i. e., the minimum mass to allow metamorphosis) as well as to a substantial reduction of the readily available feed (Bekker et al., 2021; Gligorescu et al., 2019). Chemical analyses of substrate before and after (frass) the BSFL growth are shown in supplementary material S2. Considering the different dry matter recorded for substrate and frass, a reduction of approximately 50% of total volatile solids was detected. It was accompanied by a significant decline in TOC (47–53%) and NDF (15–24%) contents, while absolute amounts of ADF and ADL increased. Different from NDF, which comprises cellulose, hemicellulose and lignin (Hindrichsen et al., 2006), ADF and ADL represents the amount of only hemicellulose and lignin (Peguero et al., 2023). While cellulose is known to be easily degraded by BSFL, hemicellulose and lignin are not directly digested (Gold et al., 2018), pointing out that after 12 d of BSFL composting, no easily digestible substrate was available. On the other hand, although the insects might have had already reached a satisfactory weight for going through the metamorphosis, they still appeared far from the pupal stage, suggesting that provision of new fresh feed (re-feeding) could have reactivated the metabolism, resulting in higher final larval weight and a new CO₂ emission peak (Zhang et al., 2021).

CO₂ emissions from the sole substrates (Fig. 2A, green curves, treatments C) were roughly stable during the first week of the experiment and increased slightly only after this period. In these substrates, CO₂ emissions could mainly be associated to microbial activity (Oonincx et al., 2010) as high fungal growth was observed in every run. Microbial activity is well known for playing a crucial role in substrate modification and CO₂ emissions (Ermolaev et al., 2019). Future studies should focus on understanding the effect of microorganisms on the GHG emissions of substrate with and without BSFL.

Concerning CH₄ (Fig. 2B) and N₂O (Fig. 2C), no clear dynamics were observed for either treatments T or C. The absence of consistency on such data was also confirmed by the cross correlation, which highlighted no correlation at the inter-run level, although some positive correlations were observed within the same run (supplementary material S1). One sample-t-test (supplementary material S3) and comparisons between effective gas concentration and accuracy (supplementary material S4) of

the measuring device showed that no difference from zero were statistically observed, suggesting that BSFL did not emit any CH₄ or N₂O in our experiment. It is consistent with previous reports, which highlighted very low or absence of production of such GHGs by BSFL (Boakye-Yiadom et al., 2022; Perednia et al., 2017), which is strictly dependent on the physico-chemical characteristics and nature of the rearing substrate (Chen et al., 2019; Ermolaev et al., 2019; Sandrock et al., 2019; Pang et al., 2020a; Mertenat et al., 2019).

4.2. Cumulative gas emissions and mass balance

Cumulative gas emissions highlighted that BSFL were mainly responsible for producing CO₂, while no significant emissions of CH₄ and N₂O could be detected (Table 1). It is in accordance with previous studies that suggested very limited CH₄ and N₂O emissions, compared with direct CO₂ emissions (Lindberg et al., 2022).

Two-sample *t*-test carried out on the overall dataset detected significant difference between the system substrate + insects and the sole substrate only for CO₂ ($P = 0.001$), suggesting that BSFL were the main contributors for the CO₂ emissions of the system. An increase in CO₂ emissions due to BSFL activity was also observed in the BSFL-mediated composting process of manure when compared to classical composting (Parodi et al., 2021). On the other hand, no statistically significant differences between treatments T and C were computed for CH₄ ($P = 0.406$) and N₂O ($P = 0.428$) cumulative emissions, confirming that production of such gasses was mainly due to microbial activity on the substrate, as already stated by Oonincx et al. (2010) and Parodi et al. (2021). However, although significant differences in CO₂ emissions were detected across the entire experiment, the intra-run ANOVA showed significant differences between treatment T and treatment C only in RUN-2 ($P = 0.002$). This highlighted that other parameters, such as environmental conditions, feeding status and larval metabolic activity might have influenced the overall GHG emissions from insects. Possible explanation for the no statistical significance in runs 1 and 3 could indeed be the low number of larvae compared with the high amount of substrate used in the present study as well as the already mentioned high microbial activity. The later might also be the main explanation for the higher amount of N₂O recorded from the treatment T in RUN-1, compared to the other runs. Furthermore, it should be noted that, although not important differences were observed on the substrate composition between different runs (supplementary material S2), the insects reared on RUN-1 showed lower growth and slightly higher survival (Table 2). These two aspects together might have affected the substrate turning activity, resulting in different air circulation within the bulk and promoting the selection of specific microbial species, particularly active on denitrification and N₂O production, as already observed by Nigussie et al. (2016) in heartworm-mediated composting processes. Chemical analyses of frass (supplementary material S2) originated along the three runs support such statement. Although an increase in nitrate and nitrite from substrate to frass was observed in all the run, the amounts of both nitrate and nitrite measured in frass from the two treatments in RUN-1 were significantly lower than those measured in the frass from the other two runs. Since N₂O is primarily originates from nitrate and nitrite reduction (Thamdrup, 2012), lower concentration of these compounds in frass may suggest higher denitrification. On the other hand, it is also interesting to observe that the amount of nitrite in residuals of treatments C from RUN-1 was significantly higher than the amount of nitrite in residuals generated in runs 2 and 3. This suggests that the denitrification process and overall N₂O production are more influenced by general environmental conditions rather than by the presence and activity of BSFL. However, it should be mentioned that such phenomena may not be universally applicable as higher larval density and/or lower feeding rate could also impact the microenvironment inside the substrate (Chen et al., 2019).

CO₂ cumulative emissions computed as g of gas per kg of dry insects, ranged between 11880 and 7760 (Table 1). Despite the high variability

Table 1

Gas emissions (CO₂, N₂O, CH₄; mean ± standard error, n = 2) of the substrate + insects systems (treatments T) and sole substrate (treatments C). Values within the same run followed by different lowercase letters are statistically different (P < 0.05). Values within the same column followed by different uppercase letters are statistically different (P < 0.05).

		CO ₂		N ₂ O		CH ₄	
		experimental (g) ^a	g-gas/g-insect (DM) ^b	experimental (mg) ^a	mg-gas/g-insect (DM) ^b	experimental (mg) ^a	mg-gas/g-insect (DM) ^b
RUN-1	T	64.34 ± 3.12 ^A	9.36 ± 0.44	7.62 ± 1.14	1.13 ± 0.27	9.10 ± 0.12	1.33 ± 0.11
	C	58.79 ± 1.92 ^B	–	4.74 ± 1.47	–	7.92 ± 0.70	–
RUN-2	T	73.75 ± 1.38 ^{aA}	11.88 ± 3.79	4.41 ± 0.52	0.74 ± 0.32	4.99 ± 0.29	0.79 ± 0.23
	C	41.37 ± 0.49 ^{bB}	–	4.74 ± 0.73	–	5.91 ± 0.13	–
RUN-3	T	72.88 ± 0.27 ^A	7.76 ± 1.14	2.62 ± 0.21	0.28 ± 0.02	7.45 ± 2.13	0.76 ± 0.11
	C	35.58 ± 9.71 ^B	–	2.19 ± 0.50	–	4.80 ± 0.40	–

^a Experimental: overall gas emitted by the system from the beginning to the end of the experimental run, calculated as area under the emission curve.

^b Computed by dividing the experimental gas recovered (g) from the treatment T (substrate + insects system) by the final insect dry mass measured from the same treatment.

Table 2

Growth and bioconversion performances (mean ± standard error, n = 2) of BSFL reared on moistened chicken feed.

	BCR (%)	FCR	Assimilated feed (g)	Substrate reduction (%)	Larval survival (%)	final larval mass (mg) ^a
RUN-1	6.04 ± 0.84	8.02 ± 0.07	55.32 ± 4.78	46.87 ± 4.05	97.00 ± 0.00	93.44 ± 13.06
	6.62 ± 1.95	9.20 ± 2.44	58.26 ± 4.49	49.11 ± 3.78	85.50 ± 6.00	114.28 ± 26.17
RUN-2	8.22 ± 1.26	5.43 ± 0.33	51.69 ± 4.33	46.60 ± 3.90	95.07 ± 2.07	125.87 ± 13.51
	8.22 ± 1.26	5.43 ± 0.33	51.69 ± 4.33	46.60 ± 3.90	95.07 ± 2.07	125.87 ± 13.51

^a Fresh mass of one single larva.

observed between the three runs, which may be explained by the uncontrolled microbial contribution, as well as by different insects' performances, cumulative CO₂ production was always significantly higher than in other studies (Boakye-Yiadom et al., 2022). Several aspects including the frequency of gas measurements, the number of reared larvae, the rearing substrate and the abiotic conditions should be considered to explain such difference. For example, Parodi et al. (2020) observed that increasing the gas sampling frequency resulted in higher cumulative CO₂ detection, thereby decreasing the evaluation errors. Additionally, Pang et al. (2020b) and Chen et al. (2019) showed that CO₂ emissions were positively correlated with substrate C/N ratio and moisture, respectively, suggesting that the substrate plays an important role in determining the actual GHG emissions. Additionally, it should be mentioned that 12 h old larvae were used in the present study, while other studies usually used 3 d, 5 d, 7 d or 10 d old larvae (Boakye-Yiadom et al., 2022). As young larvae are more active in terms of metabolic performances (Gligorescu et al., 2019), it could be hypothesised that neonate larvae produced more CO₂ than older larvae. This justifies the higher emission detected in the present study and suggests that emissions from BSFL in other studies are probably underestimated. In the present study, CO₂ cumulative emissions computed per mass unit of dry insects did not seem to be strongly related to the larval growth performances, although good correlation was computed between larval gained mass and CO₂ emission (Spearman's rank correlation: $\rho = 0.89$, $P = 0.0333$). Low correlation was instead detected between substrate reduction and CO₂ emission (Spearman's rank correlation: $\rho = 0.26$, $P = 0.6583$), as well as between CO₂ emission and FCR (Spearman's rank correlation: $\rho = -0.60$, $P = 0.2417$). It confirms the findings of Chen et al. (2019), Eriksen (2022) and Laganaro et al. (2021), while contradicts Ooninx et al. (2010) and Lindberg et al. (2022), who observed an inverse correlation between FCR and CO₂ production and a direct

correlation between substrate reduction and CO₂ production.

CH₄ emissions varied between 760 and 1328 mg kg⁻¹ of dry BSFL (Table 1). Such values were within the range of published data (between 5.5 and 10066 mg kg⁻¹ of dry BSFL) (Boakye-Yiadom et al., 2022), and confirms that methane production relies more on the rearing substrate than on the BSFL. Indeed, high CH₄ emission is only observed when manure or decaying material is used (Parodi et al., 2021), while it is negligible when chicken feed is the rearing substrate (Perednia et al., 2017). However, it should be noted that our values are higher than already published values on commercial feed (Parodi et al., 2020). Such difference might be explained by the different experimental set-up applied in our study. Specifically, while Parodi et al. (2020) reared 30000 larvae on 12 kg of substrate, a very limited larvae/substrate ratio (200 larvae on 300 g of feed) was applied in our study. In such conditions, we might hypothesise that anoxic areas were generated inside the substrate, favouring the growth of methanogenic bacteria, which may have produced CH₄ (Chen et al., 2019). Similar considerations could also be valid for N₂O. Emissions of this GHG indeed usually range between 0.33 and 118 mg kg⁻¹ of dry larvae (Boakye-Yiadom et al., 2022). Significantly higher N₂O emissions were recorded in our experiment, ranging between 275 and 1130 mg kg⁻¹ of dry larvae (Table 1). The higher N₂O emissions measured in the present study might be explained by the high feeding rate (i.e. 0.6 g of dry substrate/larvae or 50 mg of dry feed/larvae/day) as higher feeding rate has been associated to higher N emissions (Nigussie et al., 2017; Rossi et al., 2023). The high variability observed in the present experiment was mainly due to the RUN-1. Indeed, as already mentioned, smaller larvae were collected in such treatments. The reduced mobility of smaller larvae within the substrate could lead to lower oxygen dissolution and higher denitrification (Thamdrup et al., 2012), which is also supported by the lower concentration of nitrate and nitrite in the frass (supplementary information 2).

Despite the differences in cumulative gas emitted across the three runs, the mass balance of C and N in substrate + insects treatments displayed an overall recovery of approximately 93% and 85–95%, respectively (Figs. 3a and 4a). Similar recovery was also observed for C in the sole substrate treatments (92–93%; Fig. 3b), while higher variability was detected for N (80–99%; Fig. 4b). Although such values are still far from 100%, based on the observations of Parodi et al. (2020) and Pang et al. (2020a) they can still be considered satisfactory. Specifically, Parodi et al. (2020) observed an overall recovery rate of 95% for C and 101% for N by using a continuous air detection system, while Pang et al. (2020a) were able to recover 87.48–97.90% of the initial C and 74.40–98.39% of the initial N, when BSFL were reared on food waste.

In the present study, the mentioned recovery rates were similarly to those of these studies (Pang et al., 2020a; Parodi et al., 2020) and the variability on gas emission quantification was lower than in Mertenat et al. (2019). This indicates that the novel gas emission chamber works and presents high capability for large-scale implementation. However, while noteworthy performances were reached for C, the result remained

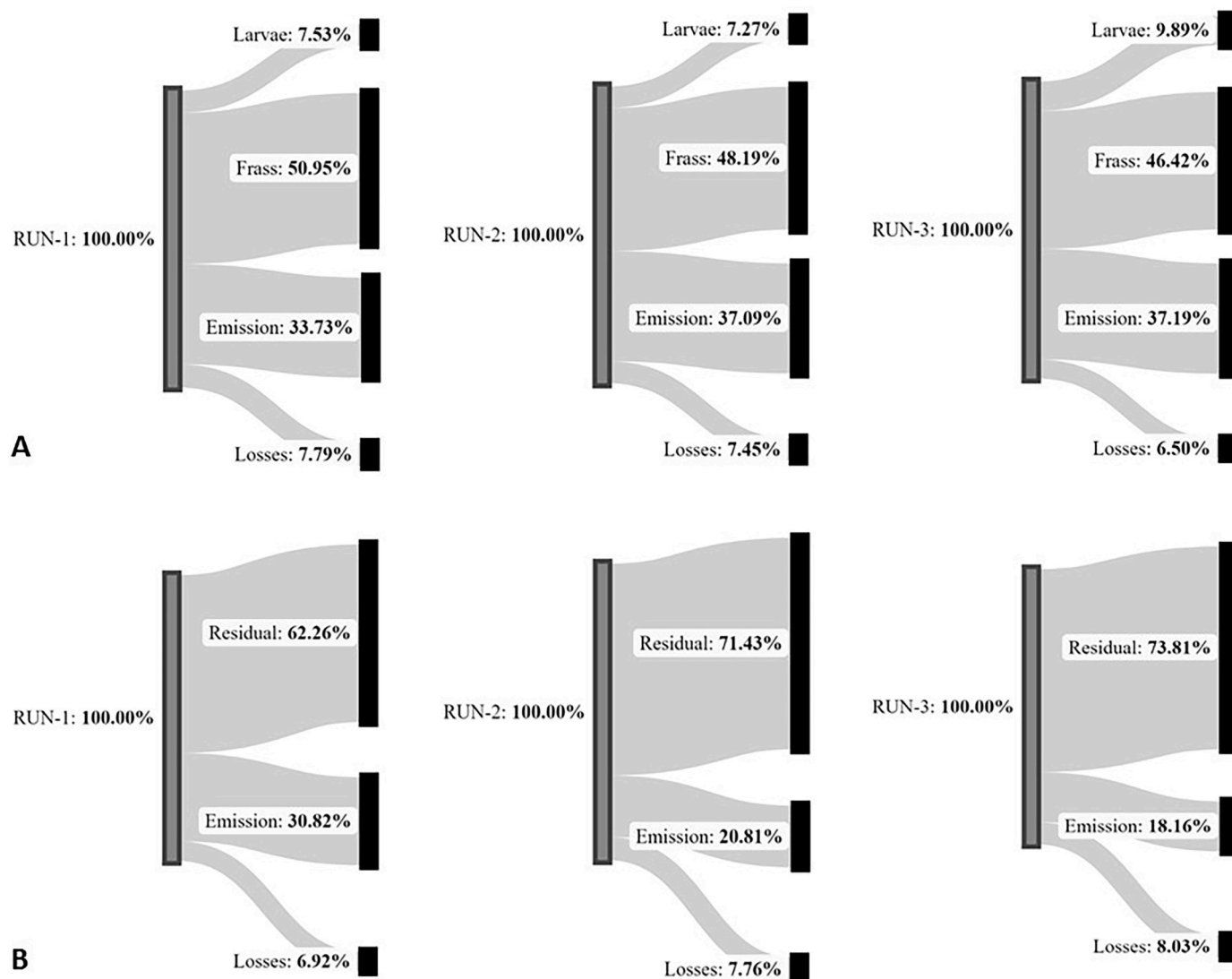


Fig. 3. Mass balance of Carbon along the three experimental runs from the substrate + insects system (A) and the sole substrate treatment (B).

low for N. The main reason for the low recovery of N might be associated to sampling frequency and accuracy of the detection method (Pang et al., 2020a), as well as the absence of a NH_3 detection system. Although NH_3 is not a direct GHG, it is often considered as main responsible for water and soil pollution (IPCC, 2014). Additionally, it might be oxidised to N_2O (FAO, 2006; Iwasa et al., 2015) contributing to increase its level in air. As NH_3 production is usually favoured by high temperature, elevated pH and increased substrate aeration (Pang et al., 2020a; Zhang et al., 2021), it can be hypothesised that significant production of such gas occurred, especially during the first days of conversion (Chen et al., 2019). It was further suggested by the strong acrid smell perceived at the end of the outlet system as well as by the significant increase of $\text{NH}_4\text{-N}$ concentration observed in both, T and C treatments (supplementary material S2). $\text{NH}_4\text{-N}$ may originate from protein degradation and mineralisation, as well as from nitrite reduction (Alvarez et al., 2014; Kroupina et al., 2022). While nitrite reduction is hardly conceivable in the studied conditions, mineralisation of protein can be hypothesised. Considering that $\text{NH}_4\text{-N}$ production is known to be responsible for increased NH_3 emissions (Yan et al., 2018; Liu et al., 2019), it is reasonable to conclude that significant NH_3 production occurred in the present experiment.

Different NH_3 production could be the reason for the high variation in the total N recovery of all treatments. Production of this gas is

extremely variable and depending on several factors, such as aeration rate, moisture, temperature and pH (Groot Koerkamp, 1994). Although detection of these factors was continuously carried out during the three experimental runs, no significant variations were observed (supplementary material S5) between the runs, excluding a potential effect of the environment on the overall NH_3 emissions. On the other hand, it has been hypothesised that different BSFL populations have different ability on producing NH_3 with not all the individuals being able to do it (Parodi et al., 2020). Another explanation could be that NH_3 emissions from BSFL rearing were strongly impacted by the larval metabolic performances (Parodi et al., 2021) as well as by the substrate microbial activity (Gold et al., 2018). Specifically, as BSFL grow, significant amounts of uric acid are produced, which are typically converted into NH_3 by microorganisms (Green and Popa, 2012).

High variability on percentage of N recovery was also observed in the treatment C (Fig. 4b) indicating different intensities of nitrification processes in the substrates. The analysed different contents of NO_2^- , NO_3^- and $\text{NH}_4\text{-N}$ in the substrates (supplementary material 2) indicate the volatilisation of a significant amount of $\text{NH}_4\text{-N}$ as NH_3 (Ren et al., 2018).

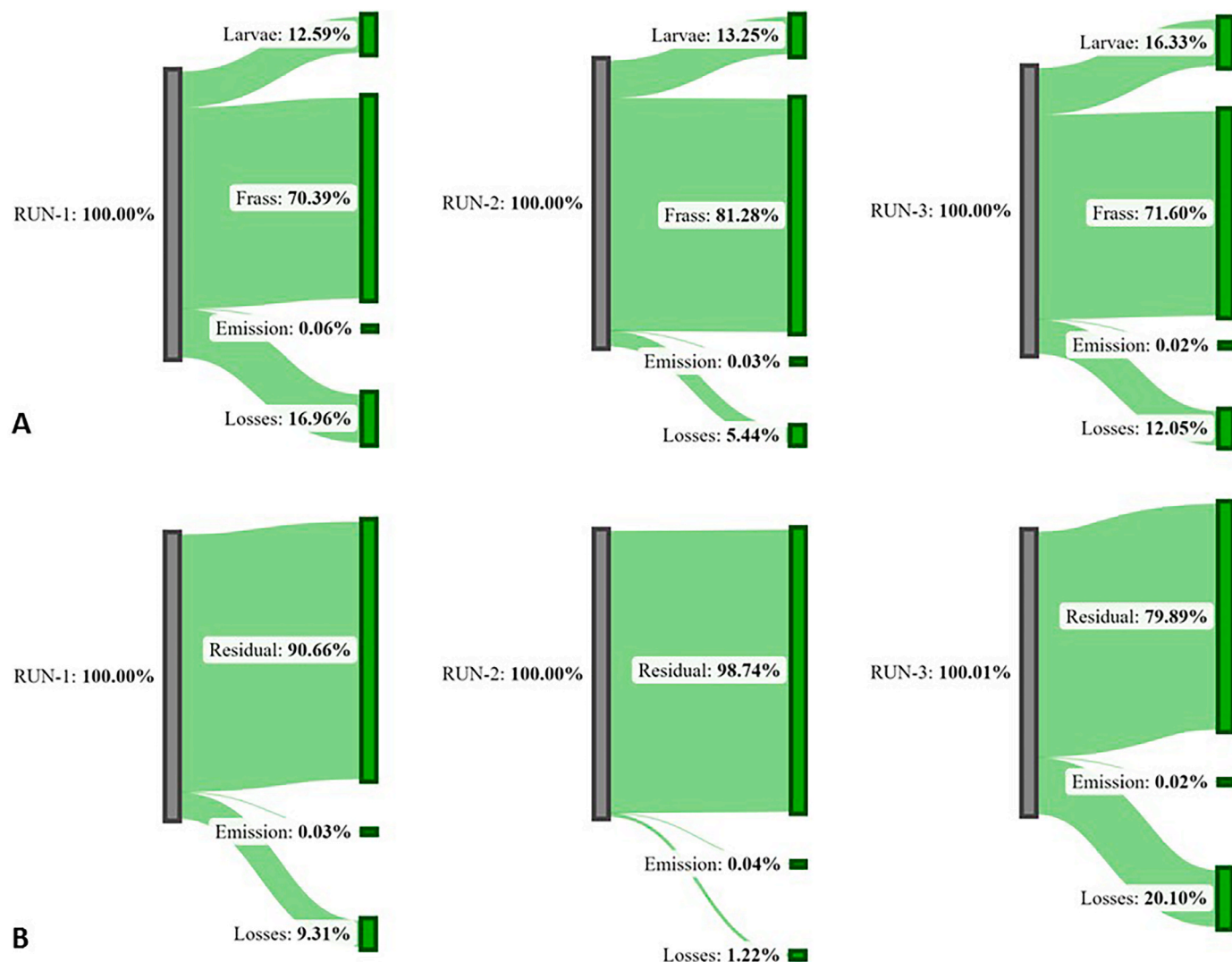


Fig. 4. Mass balance of nitrogen along the three experimental runs from the substrate + insects system (A) and the solo substrate treatment (B).

5. Conclusions

The present study showed that the novel gas emission chamber with continuous airflow is well suited for fast and reliable estimations of GHG emissions from edible insects. Although it was a first trial using the new device, gases emitted by BSFL grown on chicken feed were detected with high reproducibility in terms of CO₂ emission rate. However, the same level of reproducibility was not observed for CH₄ and N₂O emissions, which were generally limited and substrate dependent. Cumulative emissions computed as g g⁻¹ (gas/dry insect) were significantly higher than previously reported values, likely due to the use of younger larvae in this experiment. However, the absence of a continuous measurement method and the assumption that the airflow was linear between two successive measuring points might have led to overestimation of the emissions. Acceptable overall C recovery was detected for all runs, with values not significantly different from the ones reported in literature. Rather low N recovery percentage was instead detected as no NH₃ measurement was implemented.

Future studies should aim to improve the device by increasing the number of chambers in order to increase the number of parallel repetitions, implementing high-accuracy continuous emission detection systems and developing a NH₃ quantification method. Other substrates such as waste and manure, where CH₄ and N₂O emissions have already been detected, should also be tested with the aim of validate the device

for gases other than CO₂. Moreover, since the system showed low feed conversion efficiency and significant energy loss, further improvements on the experimental design, such as reduction of insects to substrate ratio, longer rearing period or repeated feeding events should be considered.

Funding

This work was financially supported by the German Federal Ministry of Food and Agriculture (BMEL) through the Federal Office for Agriculture and Food (BLE), grant number 2819ERA01MA and the BlueBio ERA-NET COFUND on the Blue Bioeconomy - Unlocking the Potential of Aquatic Bioresource and the European Commission (AquaTech4Feed project) within the Horizon 2020 program.

CRediT authorship contribution statement

Giacomo Rossi: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shikha Ojha:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Werner Berg:** Writing – review & editing, Visualization, Methodology, Investigation, Data curation. **Werner B. Herpich:** Writing – review & editing, Resources, Methodology,

Investigation, Formal analysis, Data curation, Conceptualization. **Oliver K. Schlüter**: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to thank Mr. David Sakowsky and Mr. Christian Regen for building the device, Mr. Stefan Elwert for the visual description of the device, Mrs. Miriam Felgentrau and the entire analytical chemistry group of the Leibniz Institute for Agriculture Engineering and Bioeconomy for supporting the chemical analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2024.143226>.

References

- Akbar, S.M., Pavani, T., Nagaraja, T., Sharma, H.C., 2016. Influence of CO₂ and temperature on metabolism and development of *Helicoverpa armigera* (Noctuidae: Lepidoptera). *Environ. Entomol.* 45, 229–236. <https://doi.org/10.1093/ee/nvv144>.
- Alvarez, L., Bricio, C., Blesa, A., Hidalgo, A., Berenguer, J., 2014. Transferable denitrification capability of *Thermus thermophilus*. *Appl. Environ. Microbiol.* 80 (1), 19–28. <https://doi.org/10.1128/AEM.02594-13>.
- Bekker, N.S., Heidebach, S., Vestergaard, S.Z., Nielsen, M.E., Riisgaard-Jensen, M., Zeuner, E.J., Bährndorff, S., Eriksen, N.T., 2021. Impact of substrate moisture content on growth and metabolic performance of black soldier fly larvae. *Waste Manage. (Tucson, Ariz.)* 127, 73–79. <https://doi.org/10.1016/j.wasman.2021.04.028>.
- Boakye-Yiadom, K.A., Ilari, A., Duca, D., 2022. Greenhouse gas emissions and life cycle assessment on the black soldier fly (*Hermetia illucens* L.). *Sustainability-Basel* 14. <https://doi.org/10.3390/su14161056>.
- Chen, J.S., Hou, D.J., Pang, W.C., Nowar, E.E., Tomberlin, J.K., Hu, R.G., Chen, H.C., Xie, J.S., Zhang, J.B., Yu, Z.N., Li, Q., 2019. Effect of moisture content on greenhouse gas and NH₃ emissions from pig manure converted by black soldier fly. *Sci. Total Environ.* 697 <https://doi.org/10.1016/j.scitotenv.2019.133840>.
- Diener, S., Zurbrugg, C., Tockner, K., 2009. Conversion of organic material by black soldier fly larvae: establishing optimal feeding rates. *Waste Manag. Res.* 27 (6), 603–610. <https://doi.org/10.1177/0734242X09103838>.
- Dzepe, D., Nana, P., Kuitche, H.M., Kimpara, J.M., Magatsing, O., Tchuinkam, T., Djouaka, R., 2021. Feeding strategies for small-scale rearing black soldier fly larvae (*Hermetia illucens*) as organic waste recycler. *SN Appl. Sci.* 3, 252. <https://doi.org/10.1007/s42452-020-04039-5>.
- Eriksen, N.T., 2022. Dynamic modelling of feed assimilation, growth, lipid accumulation, and CO₂ production in black soldier fly larvae. *PLoS ONE* 17 (10), e0276605. <https://doi.org/10.1371/journal.pone.0276605>.
- Ermolaev, E., Lalander, C., Vinneras, B., 2019. Greenhouse gas emissions from small-scale fly larvae composting with *Hermetia illucens*. *Waste Manage. (Tucson, Ariz.)* 96, 65–74. <https://doi.org/10.1016/j.wasman.2019.07.011>.
- FAO, 2006. *Livestock's Long Shadow: Environmental Issues and Options*. Food and Agriculture Organization of the United Nations, Rome.
- FAO, 2022. *World Food and Agriculture – Statistical Yearbook 2022*. Rome.
- Gligorescu, A., Toft, S., Hauggaard-Nielsen, H., Axelsen, J.A., Nielsen, S.A., 2019. Development, growth and metabolic rate of *Hermetia illucens* larvae. *J. Appl. Entomol.* 143, 875–881. <https://doi.org/10.1111/jen.12653>.
- Gold, M., Tomberlin, J.K., Diener, S., Zurbrugg, C., Mathys, A., 2018. Decomposition of biowaste macronutrients, microbes, and chemicals in black soldier fly larval treatment: a review. *Waste Manage. (Tucson, Ariz.)* 82, 302–318. <https://doi.org/10.1016/j.wasman.2018.10.022>.
- Green, T.R., Popa, R., 2012. Enhanced ammonia content in compost leachate processed by black soldier fly larvae. *Appl. Biochem. Biotechnol.* 166, 1381–1387. <https://doi.org/10.1007/s12010-011-9530-6>.
- Groot Koerkamp, P.W.G., 1994. Review on emissions of ammonia from housing systems for laying hens in relation to sources, processes, building design and manure handling. *J. Agric. Eng. Res.* 59, 73–87. <https://doi.org/10.1006/jaer.1994.1065>.
- Hindrichsen, I.K., Kreuzer, M., Madsen, J., Knudsen, K.E.B., 2006. Fiber and lignin analysis in concentrate, forage, and feces: detergent versus enzymatic-chemical method. *J. Dairy Sci.* 89, 2168–2176. [https://doi.org/10.3168/jds.S0022-0302\(06\)72287-1](https://doi.org/10.3168/jds.S0022-0302(06)72287-1).
- IPCC, 2014. *Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
- Iwasa, M., Moki, Y., Takahashi, J., 2015. Effects of the activity of coprophagous insects on greenhouse gas emissions from cattle dung pats and changes in amounts of nitrogen, carbon, and energy. *Environ. Entomol.* 44, 106–113. <https://doi.org/10.1093/ee/nvu023>.
- Janssen, R.H., Vincken, J.P., van den Broek, L.A., Fogliano, V., Lakemond, C.M., 2017. Nitrogen-to-protein conversion factors for three edible insects: *Tenebrio molitor*, *Alphitobius diaperinus*, and *Hermetia illucens*. *J. Agri. Food Chem.* 65 (11), 2275–2278. <https://doi.org/10.1021/acs.jafc.7b00471>.
- Kroupina, K., Bémour, C., Rose, C.F., 2022. Amino acids, ammonia, and hepatic encephalopathy. *Anal. Biochem.* 649, 114696. <https://doi.org/10.1016/j.ab.2022.114696>.
- Laganaro, M., Bährndorff, S., Eriksen, N.T., 2021. Growth and metabolic performance of black soldier fly larvae grown on low and high-quality substrates. *Waste Manage. (Tucson, Ariz.)* 121, 198–205. <https://doi.org/10.1016/j.wasman.2020.12.009>.
- Li, M.Y., Jia, N.F., Lenzen, M., Malik, A., Wei, L.Y., Jin, Y.T., Raubenheimer, D., 2022. Global food-miles account for nearly 20% of total food-systems emissions. *Nat Food* 3, 445–453. <https://doi.org/10.1038/s43016-022-00531-w>.
- Liu, T., Awasthi, M.K., Chen, H., Duan, Y., Awasthi, S.K., Zhang, Z., 2019. Performance of black soldier fly larvae (Diptera: stratiomyidae) for manure composting and production of cleaner compost. *J. Environ. Manag.* 251, 109593. <https://doi.org/10.1016/j.jenvman.2019.109593>.
- Lindberg, L., Ermolaev, E., Vinneras, B., Lalander, C., 2022. Process efficiency and greenhouse gas emissions in black soldier fly larvae composting of fruit and vegetable waste with and without pre-treatment. *J. Clean. Prod.* 338 <https://doi.org/10.1016/j.jclepro.2022.130552>.
- Linke, M., Herppich, W.B., Geyer, M., 2010. Green peduncles may indicate postharvest freshness of sweet cherries. *Postharvest Biol. Technol.* 58, 135–141. <https://doi.org/10.1016/j.postharvbio.2010.05.014>.
- Matyssek, R., Herppich, W.B., 2020. *Experimentelle pflanzenökologie*. Springer Spektrum. <https://doi.org/10.1007/978-3-662-53465-6>.
- Mazac, R., Meinila, J., Korkalo, L., Jarvio, N., Jalava, M., Tuomisto, H.L., 2022. Incorporation of novel foods in European diets can reduce global warming potential, water use and land use by over 80%. *Nat Food* 3, 286–293. <https://doi.org/10.1038/s43016-022-00489-9>.
- Mertenat, A., Diener, S., Zurbrugg, C., 2019. Black soldier fly biowaste treatment - assessment of global warming potential. *Waste Manage. (Tucson, Ariz.)* 84, 173–181. <https://doi.org/10.1016/j.wasman.2018.11.040>.
- Moruzzo, R., Mancini, S., Guidi, A., 2021. Edible insects and sustainable development goals. *Insects* 12 (6), 557. <https://doi.org/10.3390/insects12060557>.
- Nigussie, A., Bruun, S., de Neergaard, A., Kuyper, T.W., 2017. Earthworms change the quantity and composition of dissolved organic carbon and reduce greenhouse gas emissions during composting. *Waste Manage. (Tucson, Ariz.)* 62, 43–51. <https://doi.org/10.1016/j.wasman.2017.02.009>.
- Nigussie, A., Kuyper, T.W., Bruun, S., de Neergaard, A., 2016. Vermicomposting as a technology for reducing nitrogen losses and greenhouse gas emissions from small-scale composting. *J. Clean. Prod.* 139, 429–439. <https://doi.org/10.1016/j.jclepro.2016.08.058>.
- Oonincx, D.G.A.B., van Itterbeeck, J., Heetkamp, M.J.W., van den Brand, H., van Loon, J. J.A., van Huis, A., 2010. An exploration on greenhouse gas and ammonia production by insect species suitable for animal or human consumption. *PLoS One* 5 (12), e14445. <https://doi.org/10.1371/journal.pone.0014445>.
- Pang, W.C., Hou, D.J., Chen, J.S., Nowar, E.E., Li, Z.T., Hu, R.G., Tomberlin, J.K., Yu, Z. N., Li, Q., Wang, S.C., 2020a. Reducing greenhouse gas emissions and enhancing carbon and nitrogen conversion in food wastes by the black soldier fly. *J. Environ. Manag.* 260 <https://doi.org/10.1016/j.jenvman.2020.110066>.
- Pang, W.C., Hou, D.J., Nowar, E.E., Chen, H.C., Zhang, J.B., Zhang, G.P., Li, Q., Wang, S. C., 2020b. The influence on carbon, nitrogen recycling, and greenhouse gas emissions under different C/N ratios by black soldier fly. *Environ. Sci. Pollut. Res.* 27, 42767–42777. <https://doi.org/10.1007/s11356-020-09909-4>.
- Parodi, A., De Boer, I.J.M., Gerrits, W.J.J., Van Loon, J.J.A., Heetkamp, M.J.W., Van Schelt, J., Bolhuis, J.E., Van Zanten, H.H., 2020. Bioconversion efficiencies, greenhouse gas and ammonia emissions during black soldier fly rearing - a mass balance approach. *J. Clean. Prod.* 271 <https://doi.org/10.1016/j.jclepro.2020.122488>.
- Parodi, A., Gerrits, W.J.J., Van Loon, J.J.A., De Boer, I.J.M., Aarnink, A.J.A., Van Zanten, H.H.E., 2021. Black soldier fly reared on pig manure: bioconversion efficiencies, nutrients in the residual material, greenhouse gas and ammonia emissions. *Waste Manage. (Tucson, Ariz.)* 126, 674–683. <https://doi.org/10.1016/j.wasman.2021.04.001>.
- Parodi, A., Leip, A., De Boer, I.J.M., Slegers, P.M., Ziegler, F., Temme, E.H.M., Herrero, M., Tuomisto, H., Valin, H., Van Middelaar, C.E., Van Loon, J.J.A., Van Zanten, H.H.E., 2018. The potential of future foods for sustainable and healthy diets. *Nat. Sustain.* 1, 782–789. <https://doi.org/10.1038/s41893-018-0189-7>.
- Peguero, D.A., Gold, M., Endara, A., Niu, M.T., Zurbrugg, C., Mathys, A., 2023. Evaluation of ammonia pretreatment of four fibrous biowastes and its effect on black soldier fly larvae rearing performance. *Waste Manage. (Tucson, Ariz.)* 160, 123–134. <https://doi.org/10.1016/j.wasman.2023.01.033>.

- Perednia, D.A., Anderson, J., Rice, A., 2017. A comparison of the greenhouse gas production of black soldier fly larvae versus aerobic microbial decomposition of an organic feed material. *Res. Rev.: J. Ecol. Environ. Sci.* 5.
- Poore, J., Nemecek, T., 2018. Reducing food's environmental impacts through producers and consumers. *Science* 360, 987–992. <https://doi.org/10.1126/science.aag0216>.
- Ren, X., Awasthi, M.K., Wang, Q., Zhao, J., Li, R., Tu, Z., Chen, H., Awasthi, S.K., Zhang, Z., 2018. New insight of tertiary-amine modified bentonite amendment on the nitrogen transformation and volatile fatty acids during the chicken manure composting. *Bioresour. Technol.* 266, 524–531. <https://doi.org/10.1016/j.biortech.2018.07.010>.
- Rossi, G., Ojha, S., Müller-Belecke, A., Schlüter, O.K., 2023. Fresh aquaculture sludge management with black soldier fly (*Hermetia illucens* L.) larvae: investigation on bioconversion performances. *Sci. Rep.* 13 (1), 20982 <https://doi.org/10.1038/s41598-023-48061-0>.
- Rux, G., Caleb, O.J., Frohling, A., Herppich, W.B., Mahajan, P.V., 2017. Respiration and storage quality of fresh-cut apple slices immersed in sugar syrup and orange juice. *Food Bioprocess Technol.* 10, 2081–2091. <https://doi.org/10.1007/s11947-017-1980-6>.
- Sandrock, C., Walter, C., Wohlfahrt, J., Krauss, M., Amelchanka, S., Berard, J., Leiber, F., Kreuzer, M., 2019. Greenhouse gas emissions of black soldier fly larvae grown on different feed substrates throughout larval development. *Bornimer Agrartechnische Berichte* 79.
- Sripem, N., Pesti, G.M., Tillman, P.B., 2011. Evaluation of the fixed nitrogen-to-protein (N:P) conversion factor (6.25) versus ingredient specific N: P conversion factors in feedstuffs. *J. Sci. Food Agric.* 91, 1182–1186. <https://doi.org/10.1002/jsfa.4292>.
- Thamdrup, B., 2012. New pathways and processes in the global nitrogen cycle. *Annu. Rev. Ecol. Evol. Systemat.* 43, 407–428. <https://doi.org/10.1146/annurev-ecolsys-102710-145048>.
- Yan, L., Wang, G., Ai, S., Huo, Z., Wang, Y., Gu, J.D., Wang, W., 2018. Abundance of ammonia-oxidizing bacteria and archaea under different ventilation strategies during cattle manure composting. *J. Environ. Manag.* 212, 375–383. <https://doi.org/10.1016/j.jenvman.2018.02.032>.
- Zhang, X.Y., Li, Z.T., Nowar, E.E., Chen, J.S., Pang, W.C., Hou, D.J., Hu, R.G., Jiang, H., Zhang, J.B., Li, Q., 2021. Effect of batch feeding times on greenhouse gas and NH₃ emissions during meat and bone meal bioconversion by black soldier fly larvae. *Waste Biomass Valor* 12, 3889–3897. <https://doi.org/10.1007/s12649-020-01277-x>.